Title: Sparsity in Quantum Simulations

In quantum computing, simulation methods encounter memory-intensive challenges, exponentially increasing memory requirements with each additional qubit. In this study, we investigate the sparsity patterns observed in quantum simulations using state-vector and tensor-network approaches. Leveraging these patterns, we introduce a specialized data structure for storing matrices in state-vector simulations and tensors in tensor-network simulations. This approach significantly reduces the memory footprint and simulation time through a set of tailored kernels for tensor operations on the data structure.

To demonstrate the viability of our approach, we employ the supermarQ benchmark suite, which includes quantum circuits such as GHZ, Hamiltonian, and Mermin-Bell pair generation. Remarkably, the identified sparsity patterns also manifest in real-life quantum algorithms such as Grover and Max-Cut (VQE). Initially, we empirically establish the advantages of our data structure for quantum simulations by counting the number of floating-point operations (FLOPS) in chain matrix multiplication, mimicking a state-vector simulation.

Subsequently, we delve into tensor-network-based simulations and discover that higher-dimensional tensors within tensor networks exhibit similar sparsity patterns when reshaped and stored as matrices in a certain fashion. Capitalizing on this finding, we develop specialized kernels for tensor operations like reshape, transpose and tensordot using this novel data structure. We encapsulate these computational functionalities into a numerical library, which is made accessible as a Python package using pybind11 bindings. Furthermore, we seamlessly integrate this library with Quimb, facilitating tensor-network simulations.

Our findings reveal that the main computational bottlenecks reside in format conversions and transpose operations. While the current simulation times are relatively longer, our approach enables simulations with a greater number of qubits before requiring distributed computing beyond a single node. In contrast, existing non-sparse libraries employing different data structures tend to exhaust memory resources for a smaller number of qubits.

In summary, this work introduces a novel data structure that leverages sparsity patterns in quantum simulations, resulting in a reduced memory footprint and improved simulation times. The demonstrated benefits highlight the potential of our approach to enhance the efficiency and scalability of quantum algorithms in practical implementations.